Infolink University College – Hawassa Campus
Department of Computer Science
Guideline for Software Engineering

Real life project is an essential part of the university curriculum for the students of B.Sc Computer Science and IT to give them soul sentity of the technology. Following guidelines are designed for the students of Computer Science and IT for their **Software Engineering** . It will serve for uniformity and consistency in project proposals and main project writing by the target students.

1. Format or writing style in Software Engineering Project

Paper size and margins

- Use A-4 paper (8 1/2 x 11") and 2.5 cm for all margins of the manuscripts

Line and paragraph spacing

- Use 1.5 spacing for the body of the text, except for tables and references, where you need to use single line spacing. Do not indent paragraphs but use block typing and no need of background effects. Alignment of the text is essential.

Font type and font size

- Capitalize only the first letter of each word, excluding common words in the title and make its font 16 and Bold. The common words are prepositions, conjunctions or connectives (such as: of, in, a, and, or, etc.)

Example 1: Title & Font size

Your Title of The Project (write the title here)

- Capitalize only the first letter of the main heading and make its font size 16 and bold as above.

Example 2: Subheading and Font size

Chapter One

Introduction

- Capitalize only the first letter of the subheading and make its font size 14 and bold as above.

Example 3: Sub-subheading

Network Security

- If there is a sub-subheading, capitalize only the first letter and make it italic with a font size of 12 without bolding as above.

2. References or Bibliography , Webliography

Use the following format (APA citation style)

Author(s) (date & year), Title of Book, Title of Article, Title of Periodical, Volume, Pages, Place of Publication, Publisher and Other Information.

Example:

References:

James, N. E. (1988). *Two sides of paradise: The Eden myth according to Kirk and Spock*. In D. Palumbo (Ed.), Spectrum of the fantastic (pp. 219-223). Westport, CT: Greenwood.

Lynch, T. (1996). *DS9 trials and tribble actions review*. Retrieved March 08, 2010, from Psi Phi: Bradley's Science Fiction Club

INFOLINK COLLEGE HAWASSA CAMPUS

"Title BOLD"

Group members name and Id no.

Advisor's name

Documentation (Proposal) of Course Software Engineering

Submitted to Department of Computer Science, Faculty of Engineering, IC, Infolink College, in Partial fulfillment for the requirement of the Degree of Bachelor Science in (Computer Science)

Hawassa, Ethiopia
March, 2025

4. Contents of senior project proposal

Senior Project Proposal

Since some parts of this proposal will be included in your final project document, you are expected to do your best. The project proposal shall have the following sections:

1. **Title page (see its format)**
2. **Abstract (approximately 200 to 400 words with keywords)**

Provide a brief summary of your project. A few sentences should suffice since you will provide details in other sections.

3. **Table of content**
4. **List of tables (if any)**
5. **Background of The company**
6. **Background of the Project**

Provide information essential to understanding your project. This includes, but is not limited to, the following:

- Descriptions (Provide brief description about institute/university, details of your project, why you select the topic? Major goals of the project, Why this topic is of interest)
  - Prior work done by others (if any)
  - Prior work done by you (if any)

7. **Team Composition**

Project Title	Full name of the project title (short name)
Prepared By	S.No. Name ID. No. Email/Mobile Responsibility
Date	…… March, 2010
Advisor	

8. **Tasks and Schedule (**List the start date, stop date, and estimated number of hours to completion. Your schedule shall include submitting the requirements specification as a milestone. Be sure to consider holidays and other events that may impact your schedule. Be realistic!)

9. **Problem statement**

10. **Detailed Objectives of the Project**
    1.
    2.
    3.
    .
    .

These objectives will be achieved by following the implementation through successive activities:

- Problem identification and definition
- Requirement Gathering
- System analysis
- System design
- Software development/Customization
- Testing
- Pre implementation
- Rectification of desired parameters
- Post implementation
- Documentation

11. **Scope of the Project**

12. **Feasibility (Optional)**

- Technical Feasibility
- Operational Feasibility
- Political/Behaviourable Feasibility
- Schedule Feasibility

13. **Methodology**

- Data Source
- Fact-finding Techniques
- Analysis and Design Approach
- Overview of Project Phases
- Artifacts to Produce
  I. Inception phase
  II. Elaboration Phase (SRS-System Requirement Specifications)
  III. Construction Phase
  IV. Transition Phase

- **Development Tools (sample is given below)**

Activities	Tools/ Programs
Client side coding	*HTML/DHTML/XML*
Client side scripting	*JavaScript*
Platform	*MS Windows or Linux*
Database server	*Mysql*
Web server	*Apache*
Server-side scripting	*Php*

Browsers	*IE 5.5/6.0/7.0, Mozilla Firefox 3.0.*
Editors	*Macromedia Dreamweaver, MS Excel,*
Documentation	*MS Word, MS Excel*
User Training	*MS PowerPoint, Video Player*
Varied technologies	*As per the technical requirement in future*

- **Required Resources with Costs (**List (and describe, as appropriate) resources needed to complete your project. This includes, but is not limited to, hardware, software and reference material. Clearly specify if you expect the university to supply any of those resources (e.g., lab computers). Specify the estimated cost for each resource.

  - **Testing Procedure (Optional)**
  - **Installation and Configuration(Optional)**
  - **Implementation(Optional)**

14. **Limitation of the Project (Optional)**
15. **References (**Provide a bibliography of reference material).
16. **Webliography (List of websites/portals)**

5. Final Report Layout of senior project

Preliminary pages

I. Title page (See its format)
II. Approval sheet
III. Dedication (optional)
IV. Acknowledgements
V. Table of contents
VI. List of figures
VII. List of Tables
VIII. Abstract
IX. Abbreviations
X. Patents Information (optional)

Chapter One: Introduction

1 Introduction
1.1 Background information of the Organization
  1.1.1 Vision of Infolink College
  1.1.2 Mission of Infolink College
1.2 Background of the project
1.3 Team composition
1.4 Statement of the problem
1.5 Objective of the project

1.5.1 General Objective
1.5.2 Specific objective

1.6 Feasibility Analysis

1.6.1 Operational feasibility
1.6.2 Technical feasibility
1.6.3 Economic feasibility
1.6.4 Behavioral/Political feasibility
1.6.5 Schedule feasibility

- Cost Benefit Analysis
- Cost of the project
- Cost break down
  - Recurrent Cost
  - One time Cost

1.7 Scope and significance of the project
1.8 Target beneficiaries of the system
1.9 Methodology for the project

1.9.1 Data Source
1.9.2 Fact Finding Techniques

- Interview
- Practical Observation
- Document Analysis

1.9.3 Systems Analysis and Design
1.9.4 Development Tools
1.9.5 Testing procedures
1.9.6 Implementation (Parallel/Partial/Direct)
1.9.7 Limitation of the project
1.9.8 Risks (What if Analysis?), Assumptions and Constraints (optional)

Chapter Two: Description of the Existing System

2.1 Introduction of Existing System
2.2 Players in the existing system
2.3 Major functions/activities in the existing system like inputs, processes & outputs
2.4 Business rules
2.5 Report generated in the existing system
2.6 Forms and other documents of the existing systems
2.7 Bottlenecks of the existing system (using for example PIECES frame Work).

2.7.1 Performance (Response time)
2.7.2 Input (Inaccurate/redundant/flexible) and Output (Inaccurate)
2.7.3 Security and Controls
2.7.4 Efficiency
2.7.5 …..If Any

2.8 Practices to be preserved
2.9 Proposed solution for the New system that address problems of the existing system (As an alternative)
2.10. Requirements of the Proposed System

2.10.1 Functional requirements

- Performance requirements
- Process requirements
- Input related requirements
- Output related requirements
- Storage related requirements

2.10.2 Non functional requirements

- Performance
- User Interface
- Security and Access permissions
- Backup and Recovery

Chapter Three: System Analysis (Modeling of the Existing and Proposed System using the chosen methodology)

3.1 Introduction
3.2 System Requirement Specifications (SRS)
3.2.1 Use case diagrams
3.2.2 Use case documentation (for each use case identified)

- Security Login
- Registration

3.2.3 Sequence diagram
3.2.4 Activity Diagram
3.2.5 Analysis level class diagram (conceptual modeling)
3.2.6 User Interface Prototyping
3.2.7 Supplementary specifications

Chapter Four: System Design

4.1 Introduction
4.2 Class type architecture

- User interface layer
- Controller/process layer
- Business/Domain layer
- Persistence layer
- System layer

4.3 Class modeling
4.4 State chart modeling
4.5 Collaboration Modeling
4.6 Component Modeling
4.7 Deployment modeling
4.8 Persistence modeling
4.9 User Interface design

Chapter Five: Implementation and Testing

5.1 Introduction
5.2 Final Testing of the system

5.3 Hardware software acquisitions
5.4 User manual preparation
5.5 Training
5.6 Installation Process
5.7 Start-up strategy

Chapter Six: Conclusions and Recommandation

6.1 Conclusions
6.2 Recommandations
- Appendix
- Références

Final deliverables:

- Documentations, both in hard copy and softcopy
- Software (on CD)

6. Project Presentation & Demonstration

As part of the assessment, students will be required to make a presentation and demonstration of their project to their assessment team/examiners.
Each presentation will be timetabled for between 30 and 40 minutes (to be announced) including questions and answers. Second marker will be part of the team but you should bear in mind that the majority of the panel will not be familiar with your project; you should take this into account when planning your presentation. Your advisors will help you to structure your talk and will be willing to go through it with you beforehand. The presentation and demonstration are assessed separately and compulsory component of the project. The assessment team will not allocate a mark for a project unless there had been a formal presentation and demonstration based on the schedule for each. The objective of the presentation is to find out exactly what you have done and to ensure that you get an accurate mark that is consistent with other projects - it is not designed as an opportunity to shoot you down!

7. Prize

The top projects recommended by examiners will be reviewed shortly after the presentations and a list of prize candidates will be drawn up. These "prize finalists" will be invited to re-present their work at a special celebration event open to the university. At the end of the day there will be a vote for a "Best Presentation" award and the departmental project prizes will be decided some time afterwards on the basis of the university wide presentations, reports and assessment team comments.

Project coordinators